RAG Basics: Production Code, MMR and Context Relevance (Answer Key)

Q1. What is the primary purpose of RAG in Generative AI?

A. To train a new LLM from scratch

B. To combine retrieval of external information with text generation

C. To reduce GPU memory usage

D. To replace vector databases

Answer: B

Q2. In a typical RAG application, the main.py file is usually responsible for:

A. Storing embeddings permanently

B. Orchestrating the overall application flow

C. Training the embedding model

D. Managing database backups

Answer: B

Q3. Which component retrieves relevant documents in a RAG pipeline?

A. Generator

B. Retriever

C. Tokenizer

D. Evaluator

Answer: B

Q4. What does MMR stand for in document retrieval?

A. Maximum Matching Retrieval

B. Minimum Model Response

C. Maximal Marginal Relevance

D. Multi-Modal Ranking

Answer: C

Q5. Why is MMR used in RAG systems?

A. To increase model size

B. To retrieve diverse yet relevant documents

C. To reduce tokenization time

D. To generate embeddings

Answer: B

Q6. If the top 5 retrieved documents are very similar to each other, which technique can help improve diversity?

A. Fine-tuning

B. MMR

C. Quantization

D. Caching

Answer: B

Q7. What is Context Relevance in a RAG system?

A. How fast the response is generated

B. How relevant the retrieved context is to the user's query

C. The number of tokens in a document

D. The size of the vector database

Answer: B

Q8. Why is context relevance important?

A. It helps the model generate more accurate answers

B. It reduces internet bandwidth usage

C. It increases embedding dimensions

D. It changes the model architecture

Answer: A

Q9. Which of the following is an example of poor context relevance?

A. Retrieving documents about Python for a Python question

B. Retrieving documents about cloud computing for a cloud-related question

C. Retrieving sports articles for a question about machine learning

D. Retrieving company policies for an HR query

Answer: C

Q10. In a production RAG application, what is the main benefit of separating retrieval logic into dedicated functions or modules?

A. It makes the code harder to understand

B. It improves maintainability and reusability

C. It eliminates the need for embeddings

D. It guarantees 100% accuracy

Answer: B

Q11. Which stage typically happens before the LLM generates the final answer?

A. Retrieval of relevant context

B. Model fine-tuning

C. Evaluation scoring

D. Dataset labeling

Answer: A